

Logistic Regression

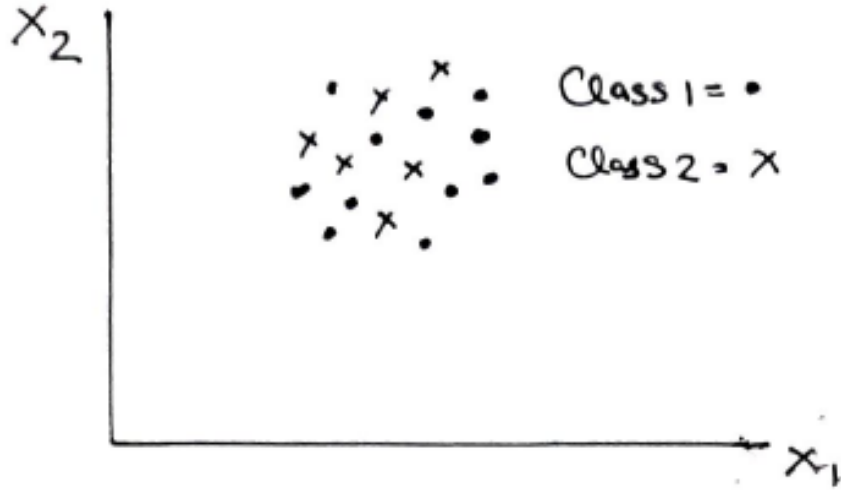


Figure 0.1: Classification

Problem setup:

1. n = Number of observations
2. p = Number of predictor variables (excluding the constant 1)
3. $y \in \{0, 1\}$ = Dependent variable (binary outcome, yes or no, qualitative)
4. $x_1, \dots, x_p$ = Independent variables (predictors)

This problem can be viewed as a classification problem where the problem is to predict which class (0 or 1) an observation belongs to or alternatively as a regression problem where the probability that the output is of a particular class (a number in the range $[0, 1]$) is predicted in terms of the independent variables.

We are interested in estimating the linear model:

$$y^* = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon.$$

However in this case, we do not observe y^* (which is a latent variable), but rather we observe a variable y which can be defined as follows:

$$y = 1 \text{ if } y^* \geq 0, \text{ and } y = 0 \text{ otherwise.}$$

Our dataset consists of $(y_i, \mathbf{x}_i)$ for $i = 1, \dots, n$ where $y_i \in \{0, 1\}$ and $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ip})$.

Key ideas:

1. The problem is to estimate probability of $y = 1$ and $y = 0$ in terms of the predictor variables x_i . Using a linear regression model is not suitable since the probability must lie between 0 and 1.

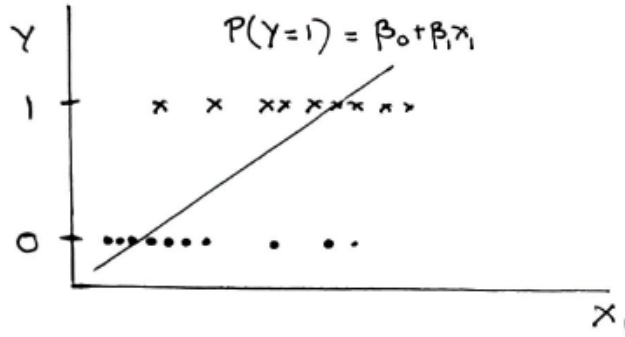


Figure 0.2: Linear regression

Define a vector of parameters:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}.$$

The logistic function (S-shaped) provides a nice formulation to capture the probability. Define:

$$P(y = 1 | \mathbf{x}, \beta) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}} = \frac{e^{\beta' \mathbf{x}}}{1 + e^{\beta' \mathbf{x}}}.$$

This number is always between 0 and 1, irrespective of the value of $\mathbf{x}$.

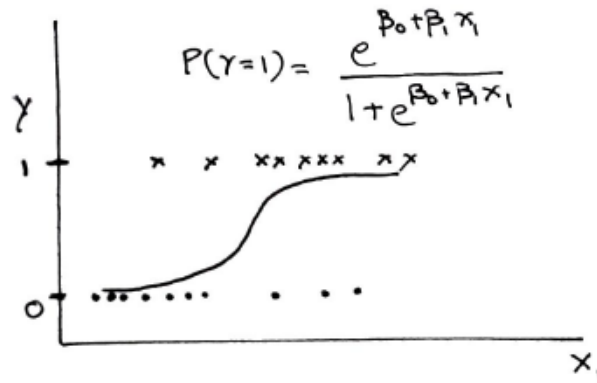


Figure 0.3: Logistic regression

2. Odds is defined as:

$$\begin{aligned}\text{Odds} &= \frac{P(y = 1|\mathbf{x}, \boldsymbol{\beta})}{P(y = 0|\mathbf{x}, \boldsymbol{\beta})} \\ &= e^{\boldsymbol{\beta}'\mathbf{x}}.\end{aligned}$$

Odds > 1 if $y = 1$ is more likely and Odds < 1 if $y = 0$ is more likely for a particular $\mathbf{x}$.

The logit or log-odds is defined as:

$$\text{Log}(\text{Odds}) = \boldsymbol{\beta}'\mathbf{x} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p,$$

and is linear in the beta coefficients. Note that a positive β_j coefficient indicates that probability of $y = 1$ increases, if x_j increases. Similarly, a negative β_j coefficient indicates that probability of $y = 1$ decreases, if x_j increases. However in contrast to linear regression, increasing the x_j variable by 1 unit (keeping all of the other x_k values fixed), changes the log-odds by β_j or multiplies the odds by e^{β_j} .

3. To estimate the coefficients, we define the likelihood function as:

$$\text{Likelihood}(\boldsymbol{\beta}) = L(\boldsymbol{\beta}) = \prod_{i=1}^n P(y = 1|\mathbf{x}_i, \boldsymbol{\beta})^{y_i} P(y = 0|\mathbf{x}_i, \boldsymbol{\beta})^{1-y_i}.$$

This computes the likelihood (probability) of seeing the actual responses that we do in the dataset for a given model with a vector of parameters, assuming that each observation is independent of the other. The maximum likelihood estimation problem is defined as:

$$\max_{\boldsymbol{\beta}} \prod_{i=1}^n P(y = 1|\mathbf{x}_i, \boldsymbol{\beta})^{y_i} P(y = 0|\mathbf{x}_i, \boldsymbol{\beta})^{1-y_i}.$$

This problem is equivalently solved by the maximum log-likelihood problem:

$$\max_{\boldsymbol{\beta}} \sum_{i=1}^n y_i \log(P(y = 1|\mathbf{x}_i, \boldsymbol{\beta})) + (1 - y_i) \log(P(y = 0|\mathbf{x}_i, \boldsymbol{\beta}))$$

where the log-likelihood is defined as:

$$\text{Log-Likelihood}(\boldsymbol{\beta}) = LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \log(P(y = 1|\mathbf{x}_i, \boldsymbol{\beta})) + (1 - y_i) \log(P(y = 0|\mathbf{x}_i, \boldsymbol{\beta})).$$

This can be reformulated as:

$$\max_{\boldsymbol{\beta}} \sum_{i=1}^n y_i (\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) - \sum_{i=1}^n \log(1 + e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}})$$

or equivalently:

$$\max_{\boldsymbol{\beta}} \underbrace{\sum_{i=1}^n y_i \boldsymbol{\beta}'\mathbf{x}_i - \sum_{i=1}^n \log(1 + e^{\boldsymbol{\beta}'\mathbf{x}_i})}_{=h(\boldsymbol{\beta})}$$

This objective function is concave in the beta variables and we are solving a maximization problem over these variables. In this case, the local optimum is the global optimum and the problem is efficiently solvable. However, we need to use an iterative algorithm to find the optimal solution in this case.

Algorithm: Define:

$$p_i(\beta' \mathbf{x}_i) = \frac{e^{\beta' \mathbf{x}_i}}{1 + e^{\beta' \mathbf{x}_i}}.$$

Let $\mathbf{y}$ denote a column vector of size n with entries y_i and $\mathbf{X}$ as the input data matrix of size $n \times (1 + p)$ with rows denoting the observations and columns denoting the intercept and the predictors. Applying Newton's method to solve this problem gives the iterative updates as:

$$\beta^{(k+1)} = \beta^{(k)} + (\nabla^2 h(\beta^{(k)}))^{-1} \nabla h(\beta^{(k)}).$$

The gradients and the Hessian matrix are computed as:

$$\nabla h(\beta) = \mathbf{X}'(\mathbf{y} - \mathbf{p}(\beta, \mathbf{X}))$$

$$\nabla^2 h(\beta) = -\mathbf{X}' \mathbf{W}(\beta, \mathbf{X}) \mathbf{X}$$

where $\mathbf{p}(\beta, \mathbf{X})$ is a vector with entries as $p_i(\beta' \mathbf{x}_i)$ and $\mathbf{W}(\beta, \mathbf{X})$ is a diagonal matrix with entries $p_i(\beta' \mathbf{x}_i)(1 - p_i(\beta' \mathbf{x}_i))$.

Quality of fit

1. Deviance is a measure of the fit of the generalized linear model (of which logistic regression is an example) where higher numbers indicates a worse fit.

Null deviance measures how well the response variable is predicted by a model that only includes the intercept.

Residual deviance measures how well the response variable is predicted by the intercept and the p predictor variables.

A significant decrease in the value from the null to residual deviance indicates that the predictor variables are useful in explaining the response variable. For logistic regression problems, suppose we estimate these coefficients as $\hat{\beta}_0$ and $\hat{\beta}$, then:

$$\text{Null deviance} = -2LL(\hat{\beta}_0),$$

where $\hat{\beta}_0$ is the estimated coefficient when only the intercept is fit and

$$\text{Residual deviance} = -2LL(\hat{\beta})$$

where $\hat{\beta}$ is the estimated coefficient vector when the intercept and the p predictor variables are fit. We can verify that when only $\hat{\beta}_0$ is fit it is chosen such that the estimated fraction of 1's from the model is exactly equal to the observed fraction of 1's in the training set.

The Akaike information criterion (AIC) is based on deviance but penalizes for making the model more complicated (similar to adjusted R-squared). However the AIC does not have a benchmark range like R-squared which lies in the interval $[0, 1]$. Smaller the AIC, the better is the model. AIC is defined as:

$$AIC = -2LL(\hat{\beta}) + 2(p + 1),$$

where the p coefficients and the intercept coefficient are estimated.

2. Likelihood ratio index (McFadden's index)

$$\rho = 1 - \frac{LL(\hat{\beta})}{LL(\hat{\beta}_0)}$$

where $\hat{\beta}$ is the estimated coefficient vector and $LL(\hat{\beta}_0)$ refers to the log-likelihood when only the intercept is fit. This compares the quality and fit of the model in comparison to a model in which only the intercept is fixed. The likelihood ratio index ranges from 0 (estimated model is no better than fitting only an intercept) to 1 (estimated model perfectly predicts the choice observed).

3. Confusion matrix: The confusion matrix is a table that allows to visualize the performance of a classification algorithm as follows.

Suppose we use the following rule to classify or predict an output:

- (a) Choose a threshold t
- (b) For any observation i with predictor variables $\mathbf{x}_i$ and estimated coefficients $\hat{\beta}$:
If $P(y = 1|\mathbf{x}_i; \hat{\beta}) \geq t$, then predict 1, else if $P(y = 1|\mathbf{x}_i; \hat{\beta}) < t$, predict 0

This results in the confusion matrix as follows:

	Actual = 0	Actual = 1
Predict = 0	True Negative (TN)	False Negative (FN)
Predict = 1	False Positive (FP)	True Positive (TP)

Table 0.1: Confusion matrix

We can define the following quantities:

$$\text{False positive rate: } \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (\text{Type I error}),$$

$$\text{True negative rate: } \text{TNR} = \frac{\text{TN}}{\text{FP} + \text{TN}} \quad (\text{Specificity}),$$

$$\text{True positive rate: } \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (\text{Sensitivity}),$$

$$\text{False negative rate: } \text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}} \quad (\text{Type II error}),$$

$$\text{Overall Accuracy } \text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{FP} + \text{TN} + \text{TP} + \text{FN}}.$$

Note that by definition, $\text{FPR} + \text{TNR} = 1$ and $\text{TPR} + \text{FNR} = 1$. By varying the threshold, we change the entries in the confusion matrix and the corresponding FPR, TPR, FNR, TNR.

4. Receiver operating characteristic (ROC) curve: Rather than computing the TPR and FPR for just a fixed threshold t , the ROC curve plots $TPR(t)$ and $FPR(t)$ as a function of t .

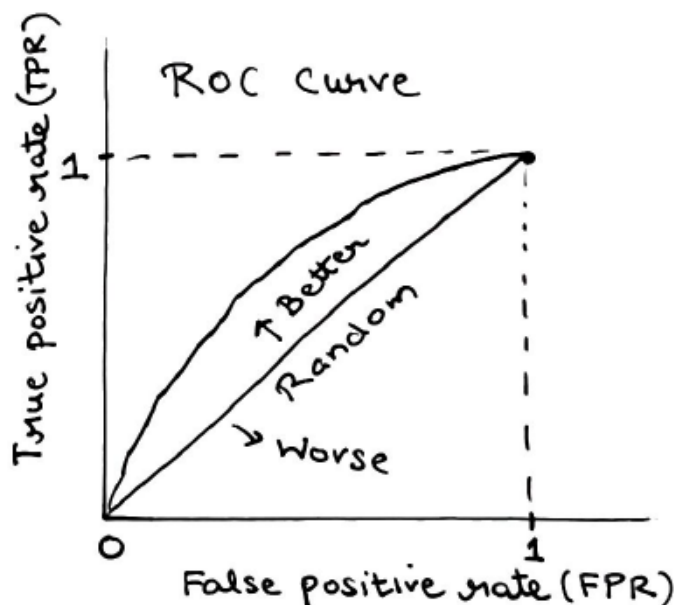


Figure 0.4: ROC Curve

- (a) Setting $t = 0$: All predictions are $y = 1$. Then $FPR = 1$ and $TPR = 1$.
 - (b) Setting $t = 1$: All predictions are $y = 0$. Then $FPR = 0$ and $TPR = 0$.
 - (c) If a model is performing at the level of chance (randomly guessing by flipping a coin), then we can achieve a point along the diagonal with $FPR = TPR$.
 - (d) A system that perfectly separates $y = 1$ from $y = 0$ has a ROC curve with $FPR = 0$ and $TPR = 1$.
5. Area under the curve: The overall performance of a classifier over all thresholds is given by the area under the ROC curve (AUC). A good predictive model has AUC closer to 1. The AUC of a classifier is the probability that a classifier will rank a randomly chosen instance with $y = 1$ higher than a randomly chosen $y = 0$ instance. A classifier that randomly guesses the class $y = 1$ half the time is expected to get half the labels with $y = 1$ and 0 correctly (the point (0.5,0.5) on the ROC curve). A classifier that randomly guesses the class $y = 1$, 90% of the time is expected to get 90% of the positives correct but FPR will also increase to 90% (the point (0.9,0.9) on the ROC curve). The AUC of a random classifier is hence 0.5.

Summary of output from logistic regression in R

Function: `glm` (use family="binomial")

1. Coefficients - This provides estimates of coefficients, standard error of coefficients, t-value and p-value ($P > |t|$). To access these use `model$coefficients` or `coefficients(model)`. For example, you can access the standard error by `coefficients(summary(model))[, "Std. Error"]`.

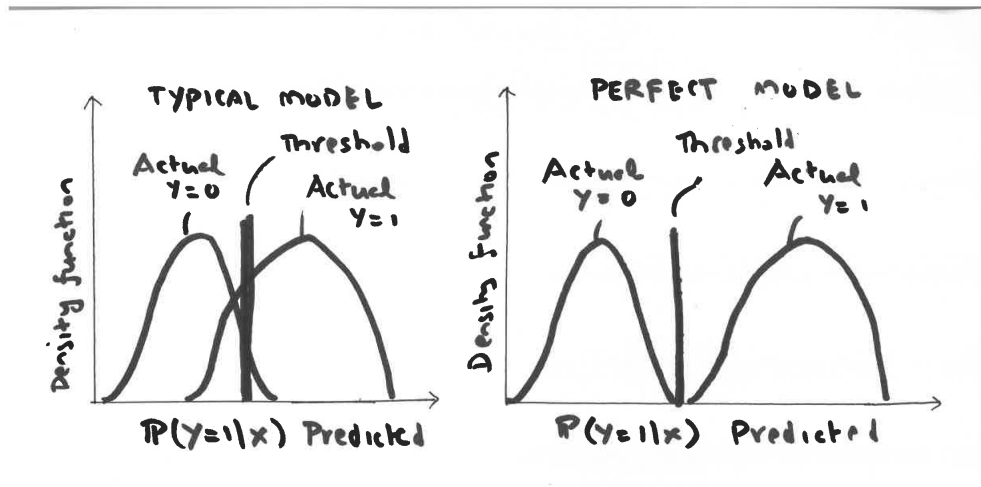


Figure 0.5: A typical model with $AUC > 0.5$ and a perfect model with $AUC = 1$

2. Null deviance, residual deviance - Null deviance and residual deviance tells you how well fit your data with the intercept only and intercept along with the other predictor variables. Smaller the values better. To access these use `model$null.deviance` and `model$deviance`.
3. AIC - Akaike information criterion given by minus twice the maximized log-likelihood plus twice the number of parameters estimated.
4. iter - Number of iterations needed.

Additional Notes

1. Concavity of log-likelihood in logistic regression:

Consider the function:

$$f(z) = \log(1 + e^z).$$

The first and second derivatives are given respectively by:

$$\frac{df}{dz} = \frac{e^z}{1 + e^z} \text{ and } \frac{d^2f}{dz^2} = \frac{e^z}{(1 + e^z)^2}.$$

Since the second derivative is nonnegative for all z , $f(z)$ is convex in z . The log-likelihood function is given by:

$$LL(\boldsymbol{\beta}) = \underbrace{\sum_{i=1}^n \underbrace{y_i \boldsymbol{\beta}' \mathbf{x}_i}_{\text{linear in } \boldsymbol{\beta}} - \sum_{i=1}^n \underbrace{\log(1 + e^{\boldsymbol{\beta}' \mathbf{x}_i})}_{\text{convex in } \boldsymbol{\beta}}}_{\text{concave in } \boldsymbol{\beta}}$$

2. Suppose we only fit the intercept. We solve:

$$\max_{\beta_0} \sum_{i=1}^n y_i \beta_0 - \sum_{i=1}^n \log(1 + e^{\beta_0}).$$

The estimate $\hat{\beta}_0$ satisfies the optimality condition:

$$\sum_{i=1}^n \left(y_i - \frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}} \right) = 0$$

or equivalently:

$$\frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}} = \frac{1}{n} \sum_{i=1}^n y_i.$$

This implies that the optimal estimate $\hat{\beta}_0$ is chosen such that the estimated fraction of ones from the model is exactly equal to the observed fraction of ones in the training set.

3. Interpretation of AUC:

AUC is the probability that a classifier, given a randomly selected instance with $y = 1$ and a randomly selected instance with $y = 0$ will be able to correctly decide which one is which.

We will discuss this interpretation in the context of logistic regression. The result generalizes to other classifiers. For any vector $\mathbf{x}$, we compute a score (value):

$$s = s(\mathbf{x}) = \frac{e^{\beta' \mathbf{x}}}{1 + e^{\beta' \mathbf{x}}}.$$

Here if the score is higher, the class will be predicted as 1 and if the score is lower, the class will be predicted as 0. Let the probability distribution function (pdf) and cumulative distribution function (cdf) of the score for class 1 be $f_1(s)$ and $F_1(s)$ while pdf and cdf of the score for class 0 be $f_0(s)$ and $F_0(s)$. We then have:

$$\text{TPR}(t) = \int_t^1 f_1(s)ds = 1 - F_1(t) \text{ and } \text{FPR}(t) = \int_t^1 f_0(s)ds = 1 - F_0(t)$$

where t is a threshold such that when the score is above t we predict 1 and 0 otherwise. We have $\text{TPR}(0) = \text{FPR}(0) = 1$ and $\text{TPR}(1) = \text{FPR}(1) = 0$. The AUC is then given by:

$$\begin{aligned} \text{AUC} &= \int_{\text{FPR}=0}^{\text{FPR}=1} \underbrace{\text{TPR}}_{\text{as function of FPR}} d\text{FPR} \\ &= \int_{t=1}^{t=0} \underbrace{\text{TPR}}_{\text{as function of } t} \text{FPR}'(t)dt \\ &= \int_{t=1}^{t=0} (1 - F_1(t))(-f_0(t))dt \\ &= \int_{t=0}^{t=1} (1 - F_1(t))f_0(t)dt \\ &= \int_{t=0}^{t=1} \left(\int_{s=t}^{s=1} f_1(s)ds \right) f_0(t)dt \\ &= \int_{t=0}^{t=1} \int_{s=0}^{s=1} 1_{s \geq t} f_1(s)f_0(t)dsdt \\ &\quad [\text{where } 1_{s \geq t} = 1 \text{ if } s \geq t \text{ and } 0 \text{ otherwise}] \\ &= \mathbb{P}(s_1 \geq s_0) \\ &\quad [\text{where } s_1 \text{ is random score for class 1 and } s_0 \text{ is random score for class 0 chosen independently}] \end{aligned}$$